Worksheet 6–2 B Riemann Sums #2 ANSWERS
AP Calculus

1.) 15750 people

2.) Since $R(t)$ is decreasing on $0 \leq t \leq 7$, the right Riemann sum is an under approximation of the integral.

3.) Approximately 15,750 people have entered the park from $t = 0$ to $t = 7$ hours since 8AM (or from 8AM to 3PM)

4.) 19125 people

5.) 36 m. The position of the bug changed approximately 36 m from $t = 0$ to $t = 12$ seconds.

6.) 5 m

7.) The average acceleration is: $\frac{v(7)-v(0)}{7-0} = \frac{4-4}{7} = 0$
Since $v(t)$ is differentiable, it is therefore continuous on $0 < t < 7$. The MVT guarantees there is a c, $0 < \mathrm{c} < 7$, where acceleration, $v'(t) = \frac{v(7)-v(0)}{7-0} = 0$.

8.) a) $1.5\ \frac{\text{gal/min}}{\text{min}}$

b.) Total consumed after 90 minutes: 84 + 4125 = 4209 gallons

9.) $\frac{1762}{105} = 16.781$

10.) The distance between two friends playing tag increased by 1.2 feet from $t = 2$ to $t = 11$ seconds

11.) The number of fans in the stands decreased by 32 from $t = 1$ to $t = 7$ seconds since the game ended.

12.) 527 students entered the school from $t = 0$ to $t = 8$ minutes after 8 AM (or from 8:00 AM to 8:08 AM).